the *curse of dimensionality* because the latter only allows to partition the search space along one direction at the time.

R-trees also use bounding envelops, but with a different shape, hyper-rectangles instead of hyper-spheres. While hyper-spheres can be stored more efficiently and allow for faster computation of their exact distance, hyper-rectangles can grow asymmetrically in different directions: this allow to cover a node with a smaller volume, while hyper-spheres, being symmetrical in all directions, generally do waste more space, with large regions without any point. And indeed, if you compare figure 10.4 to figure 10.8, you can see that rectangular bounding envelopes are much tighter to data than the spherical ones of the Ss-tree.

On the other hand, it can be proved that spheres the decomposition in spherical regions minimizes the number of leaves traversed[19]

If we compare the growth of volume of spheres and cubes in a k-dimensional spaces, for different values of k, given by these formulas:

$$V_{Cube} = r^k, V_{Sphere} = \frac{r^k \pi^{k/2}}{(k/2)!}$$

we can see that spheres grows more slowly than cubes, as also shown in the charts in figure 10.25.

And, if a group of points is uniformly distributed along all directions, and shaped as spherical cluster, then a hyper-sphere is the type of bounding envelope that wastes the